WORKED SOLUTION

STEP I 2012, Question 4

Pure • Coordinate geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q4

The key idea

Everything follows from the gradient of $y = \frac{1}{2x}$ and some careful elimination. When we intersect two lines that differ only by swapping p and q, subtracting the equations always produces a factor $p - q$, which we may cancel because the points are distinct. In the special case, it pays to notice what the condition does to Q: it makes Q the mirror image of P in the line $y = x$, and that symmetry explains why T and N land on that line.

Solution

The point T

On C we have $y = \dfrac{1}{2x}$, so $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{1}{2x^2}$. The tangent at P therefore has gradient $-\frac{1}{2p^2}$ and equation

$$y - \frac{1}{2p} = -\frac{1}{2p^2}(x - p).$$

Multiplying by $2p^2$ gives $2p^2y - p = -x + p$, that is,

$$x + 2p^2y = 2p. \tag{1}$$

In the same way the tangent at Q is

$$x + 2q^2y = 2q. \tag{2}$$

Subtracting (2) from (1),

$$2(p^2 - q^2)\,y = 2(p - q).$$

The points are distinct, so $p \neq q$, and we may divide by $2(p - q)$ to get $(p + q)y = 1$. Since $p, q > 0$, $p + q \neq 0$, so

$$y = \frac{1}{p + q}.$$

Substituting into (1),

$$x = 2p - \frac{2p^2}{p+q} = \frac{2p(p+q) - 2p^2}{p+q} = \frac{2pq}{p+q}.$$

So T is the point $\left(\frac{2pq}{p+q}, \frac{1}{p+q}\right)$, as required.

The point N when $pq = \frac{1}{2}$

The normal at P is perpendicular to the tangent, so its gradient is $2p^2$, and its equation is

$$y = 2p^2x - 2p^3 + \frac{1}{2p}. \tag{3}$$

Likewise the normal at Q is $y = 2q^2x - 2q^3 + \frac{1}{2q}$. Equating the two expressions for y:

$$2(p^2 - q^2)\,x = 2(p^3 - q^3) - \left(\frac{1}{2q} - \frac{1}{2p}\right) = 2(p^3 - q^3) + \frac{p-q}{2pq}.$$

Dividing by $2(p-q) \neq 0$,

$$(p+q)\,x = p^2 + pq + q^2 + \frac{1}{4pq}.$$

This holds for any distinct p and q. Now we use $pq = \frac{1}{2}$, so $\frac{1}{4pq} = \frac{1}{2} = pq$, and the right-hand side becomes

$$p^2 + 2pq + q^2 = (p+q)^2.$$

Hence $x = p + q$. Putting this into (3),

$$y = 2p^2(p+q) - 2p^3 + \frac{1}{2p} = 2p^2q + \frac{1}{2p} = p\,(2pq) + q = p + q,$$

where we used $2pq = 1$ twice: $2p^2q = p(2pq) = p$, and $\frac{1}{2p} = q$. Therefore

$$N = (p+q,\ p+q).$$

(Note that $p \neq q$ together with $pq = \frac{1}{2}$ means $p \neq \frac{1}{\sqrt{2}}$, which is the only restriction on p.)

Ask yourself Why should T and N lie on $y = x$ at all?

When $pq = \frac{1}{2}$ we have $\frac{1}{2q} = p$ and $\frac{1}{2p} = q$, so $P = (p, q)$ and $Q = (q, p)$. The curve $xy = \frac{1}{2}$ is symmetric under reflection in $y = x$, and this reflection swaps P and Q. It therefore swaps the two tangents and swaps the two normals, so each intersection point is mapped to itself, and a point fixed by this reflection lies on $y = x$. The algebra confirms it.

T and N on $y = x$, and the product of distances

With $2pq = 1$, the result for T gives

$$T = \left(\frac{1}{p+q}, \frac{1}{p+q} \right),$$

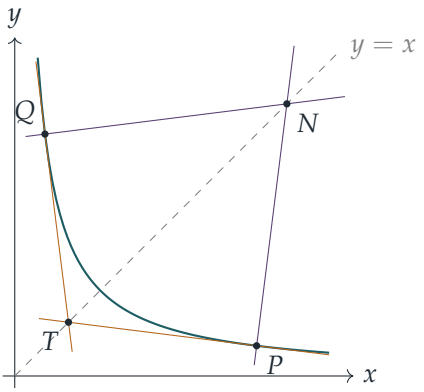
which lies on $y = x$, and we found $N = (p+q,\ p+q)$, which also lies on $y = x$.
A point (a, a) with $a > 0$ is at distance $\sqrt{2}\,a$ from the origin. Since $p + q > 0$,

$$OT = \frac{\sqrt{2}}{p+q}, \qquad ON = \sqrt{2}\,(p+q),$$

so

$$OT \cdot ON = 2,$$

which is constant (independent of p), as required.



The figure shows the case $p = 2$, $q = \frac{1}{4}$: tangents in amber, normals in plum.

Checks

Take $p = 2$, $q = \frac{1}{4}$, so $pq = \frac{1}{2}$. The tangents are $x + 8y = 4$ and $8x + y = 4$, which meet at $\left(\frac{4}{9}, \frac{4}{9}\right)$, matching $\frac{1}{p+q} = \frac{4}{9}$. The normals are $y = 8x - \frac{63}{4}$ and $y = \frac{x}{8} + \frac{63}{32}$, which meet at $\left(\frac{9}{4}, \frac{9}{4}\right)$, matching $p + q = \frac{9}{4}$. Then $OT \cdot ON = \frac{4\sqrt{2}}{9} \cdot \frac{9\sqrt{2}}{4} = 2$. We also confirmed the general formula for T, and both results in the special case, with exact rational arithmetic for 2000 random choices of p and q.

Common slips

- Dividing by $p - q$ without saying why it's allowed: this is exactly where "distinct points" is used.

- Sign errors in $\frac{1}{2q} - \frac{1}{2p} = \frac{p-q}{2pq}$ when finding N.
- Forgetting to use $pq = \frac{1}{2}$ to simplify T before claiming it lies on $y = x$.

Beyond STEP

The rectangular hyperbola $xy = c^2$ is a conic, and the fact that $OT \cdot ON$ is constant says that T and N are *inverse points* with respect to the circle $x^2 + y^2 = 2$. Inversion in a circle is a powerful transformation in geometry and in complex analysis, where it appears as the map $z \mapsto 1/\bar{z}$. Needham's *Visual Complex Analysis* has a beautiful account of inversion and its geometric properties.

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